

TrES-2b

R-2326

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-2_110408 · created 2026-08-18T22:48:09+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2455659.888242 +/- 7.4e-05 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.145 +/- 0.03
Transit depth (Rp/Rs)^2	2.09 +/- 0.86 [%]
Orbital Inclination (inc)	82.94 +/- 1.07
Airmass coefficient 1 (a1)	1.18 +/- 0.0098
Airmass coefficient 2 (a2)	-0.1271 +/- 0.0034
Scatter in the residuals of the lightcurve fit is	0.83 %
Best Comparison Star	None
Optimal Aperture	4.74
Optimal Annulus	12.0
Transit Duration (day)	0.055 +/- 0.027

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.145 $\pm$ 0.03 vs 0.1254 (deviation 0.62 σ)
Duration fit vs published	0.055 d vs 0.0746 d (deviation 0.68 σ)
Mid-time O–C	0.2 min
Depth SNR	4.6
Residual scatter	0.83 %

Light curve

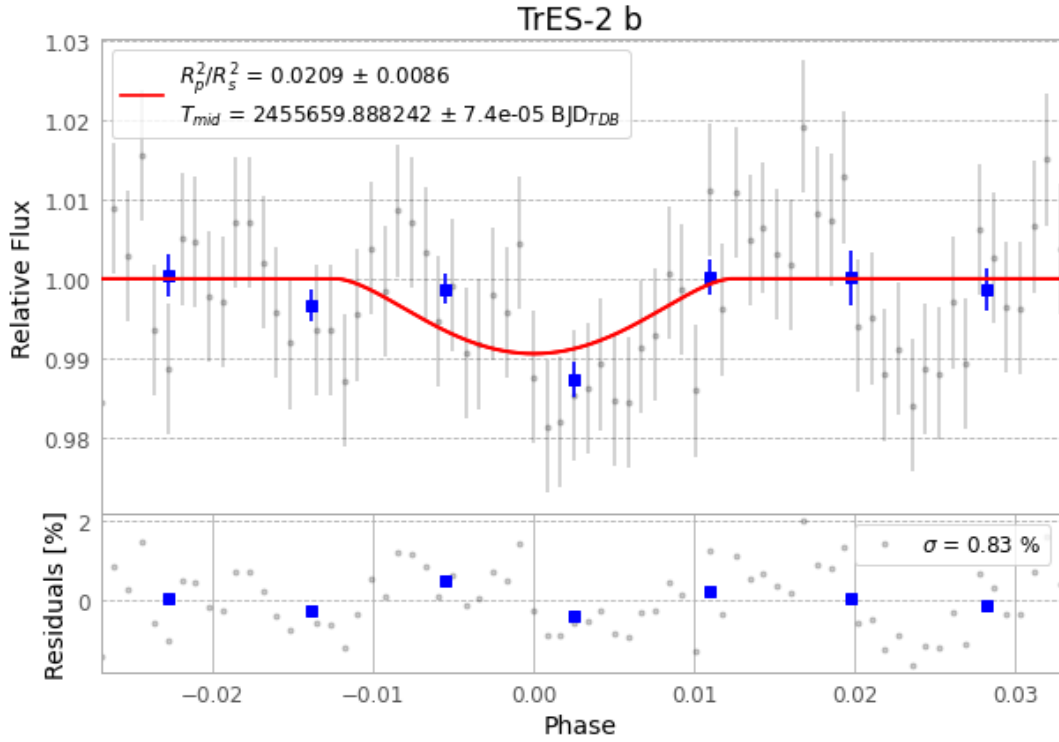


Figure 1 — FinalLightCurve_TrES-2 b_2011-04-08.png (SHA-256 567681726ac5ec60...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [296, 203], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 ced8277911a1cd48...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

72 FITS frames (47 MB), TRES-2110408074213.FITS → TRES-2110408111513.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-2-110408.log (bcf4d1c8cb2f...), FinalLightCurve_TrES-2 b_2011-04-08.png (567681726ac5...), FinalLightCurve_TrES-2 b_2011-04-08.csv (f9bbb5211c63...)

Compute resources

Wall time 102.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-18 22:48Z.